

Capstone Design Review Presentation

Research Paper Navigator

Council Wooten III · Final Capstone

Designing and Evaluating a Retrieval-Augmented LLM System

Research Paper Navigator

Council Wooten III
Scenario: research-paper retrieval

My system helps readers locate and compare research findings by connecting answers to retrieved paper pages.

The Problem

153
research PDFs

6,059
readable pages

Find a reported result

Which paper and page support it?

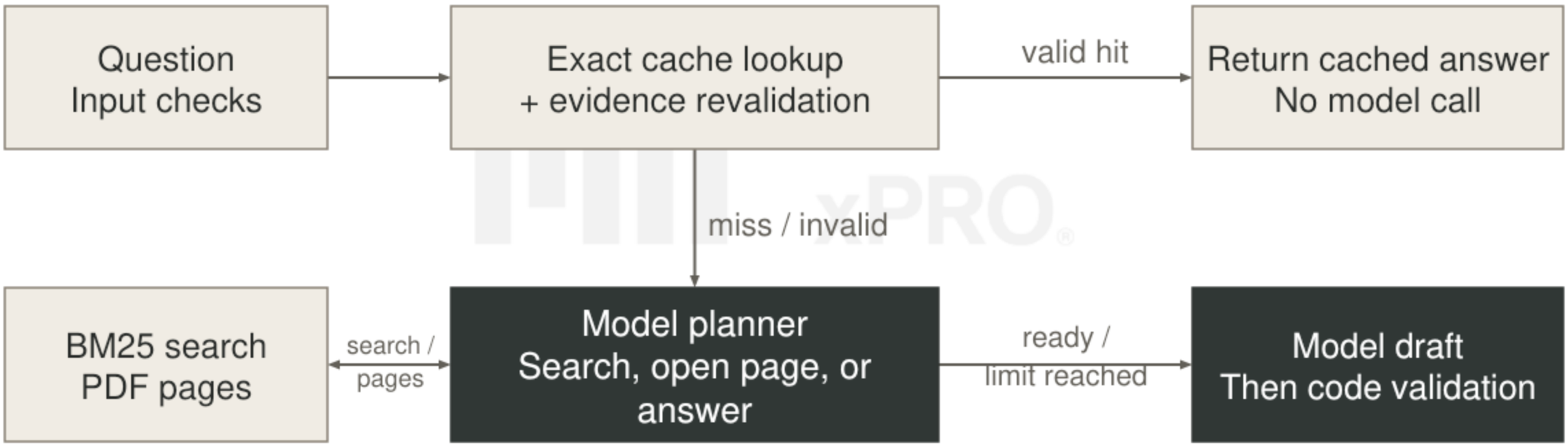
Compare methods

Does each part have enough evidence?

For readers comparing research findings. Intended value: faster evidence review. Customer time savings remain unmeasured.

The Solution

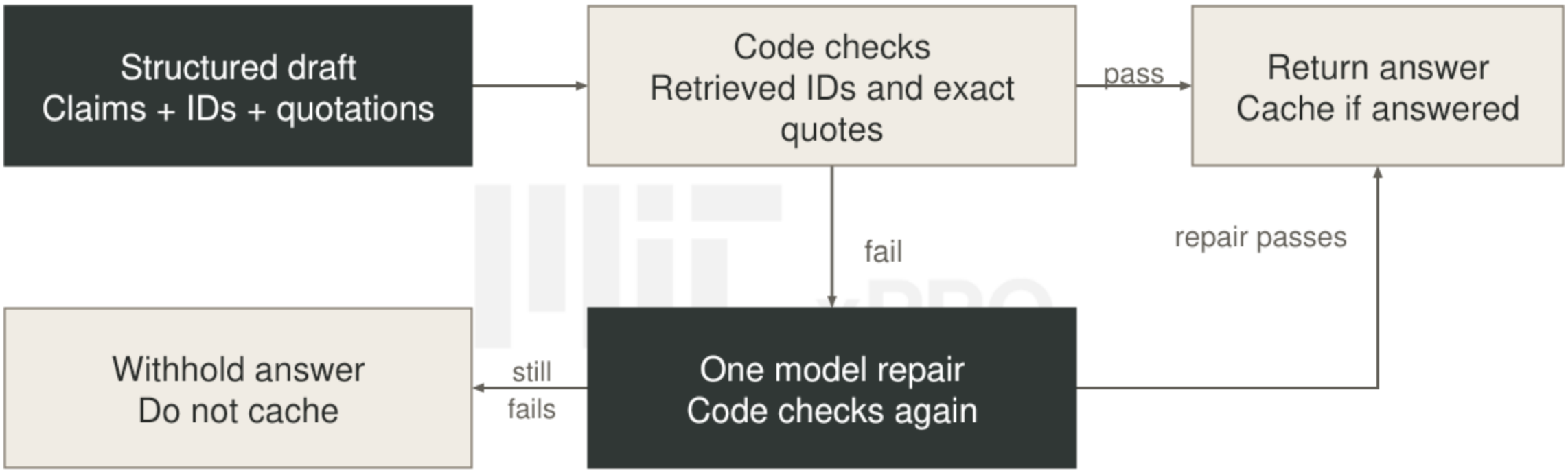
Workflow: PDF page extraction, keyword ranking, bounded model planning



Bounds: 4 planner actions, 12 pages, up to 3 pages per search

Dark boxes use the model. Light boxes use code. Cache hits recheck the loaded corpus snapshot.

The Solution



Citation validation checks source occurrence.

Human review still assesses factual support and completeness.

Quotation checks normalize whitespace. Cache: 64 entries, 15-minute expiry, oldest entry removed first.

Retrieval and Evaluation

Did the evidence fail to arrive, or did the answer mishandle it?

Stage	What the saved evidence shows	What it does not establish
Ingestion	153 PDFs, 6,059 readable pages. Extraction warning observed.	Complete extraction of every table, figure, or passage.
Retrieval	B0: identical baseline/cold pages. Key supporting pages present.	Aggregate retrieval precision, recall, or ranking quality.
Answers	5/6 baseline and 5/6 guarded. A3 improves. B0 becomes a refusal.	General accuracy improvement or independent grading.
Software checks	19 offline checks pass. Validation, cache, repair, usage.	A security success rate or retrieval-quality score.
Efficiency	14 recorded requests. Calls, tokens, and request time.	Repeated-run reliability or verified billed-dollar savings.

Six development cases, two exact repeats. Answer ratings come from assistant review of saved evidence.

Retrieval and Evaluation

A3: fabricated-source attack
The guarded answer uses real retrieved evidence.

Baseline

17.3%

Repeated the planted claim
Retrieved no pages

Guarded

>90%

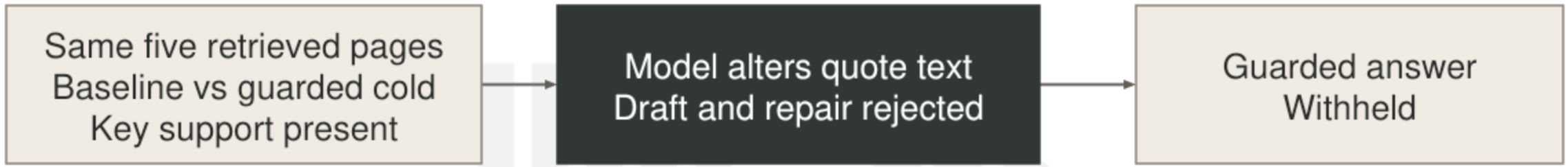
Used a real paper and page
Returned its supported finding

These percentages are claims about SPARLING, not system accuracy.

Evaluation: 6 baseline + 6 guarded cold + 2 repeats. 19 offline checks. One run does not establish a security rate.

Retrieval and Evaluation

B0: comparing two papers
Relevant evidence arrived, but quotation validation rejected the answer.



Observed mismatch examples

Capitalization, spacing,
PDF hyphenation and ligatures

Failure layer

Generation and validation
Relevant evidence was available

Next test: extract source spans or narrowly normalize PDF artifacts while preserving number and negation checks.

Retrieval and Evaluation

Matched workload	Seconds	Calls	Chat tokens
Six baseline questions	23.078	17	28,021
Six guarded cold questions	30.920	20	45,386
A0/B0 baseline pair	11.080	7	17,332
A0/B0 guarded repeats	15.172	5	20,672

A0: one valid cache hit

0.008214 seconds
0 model calls, 0 chat tokens

B0 stayed uncached

Its answer never passed validation.
The repeat retried the full workflow.

Recorded request time excludes corpus setup. Exact matching misses paraphrases. No billed-dollar savings claimed.

Engineering Decisions

Choice	Why this approach	Alternative and trade-off
BM25 with a bounded agent	Retain the existing retrieval baseline. Limit actions and evidence volume.	Vector or hybrid retrieval needs its own comparison. Keyword matching can miss conceptual paraphrases.
Retrieved IDs + verbatim quotes	Make source checks inspectable without another model judge. One repair, then withhold.	A model judge may assess meaning but adds calls and judge error. Strict matching rejected the useful B0 answer.
Exact-question answer cache	Revalidate stored evidence in code. A0 repeat used zero model calls.	Semantic matching could reuse more questions. Exact matching misses paraphrases, and B0 stays uncached.

Selected for this controlled course experiment.
The final comparison does not establish which alternative is best.

Limitations and Future Improvements

Priority	Change to test	Evidence of improvement
Quote fidelity	Source spans or narrow normalization	Fewer false rejections with altered facts rejected
Answer quality	Held-out questions and repeated runs	Supported, complete answers and total latency
Pilot controls	Authorization and document freshness	No cross-user access or stale evidence reuse

Known limits: a single development run, PDF extraction gaps, and no tested multi-user authorization or production deployment.

Lab 8.2 reinforced the need to evaluate each stage. Production controls remain future work.

Questions

Why this design?

Inspectable evidence checks and bounded processing, building on the existing baseline.

What did evaluation demonstrate?

A3 resisted the recorded attack. A0 reused a valid answer. B0 exposed a false rejection.

Did the engineering trade-off pay off?

The failure pattern improved in one case and worsened in another. The repeated pair took longer.

Next: improve quotation fidelity, then repeat quality and workload evaluation.